

CSE4227 Digital Image Processing

Chapter 03 – Sharpening Spatial Filter

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Today's Contents

❑ Sharpening spatial filter

❑ Derivatives of Image

- 1st derivative
- 2nd derivative

❑ Laplacian Filter

❑ Laplacian Image Enhancement

❑ Gradient Operators

❑ Difference filters

❑ Combining filtering techniques

■ Chapter 3 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [**Section 3.6, 3.7**]

Sharpening Spatial Filters

Previously we have looked at smoothing filters which **remove fine detail**.

Sharpening spatial filters seek to **highlight fine detail**.

- Remove blurring from images
- Highlight edges
- Useful for emphasizing transitions in image intensity

Sharpening Spatial Filters

Some Applications

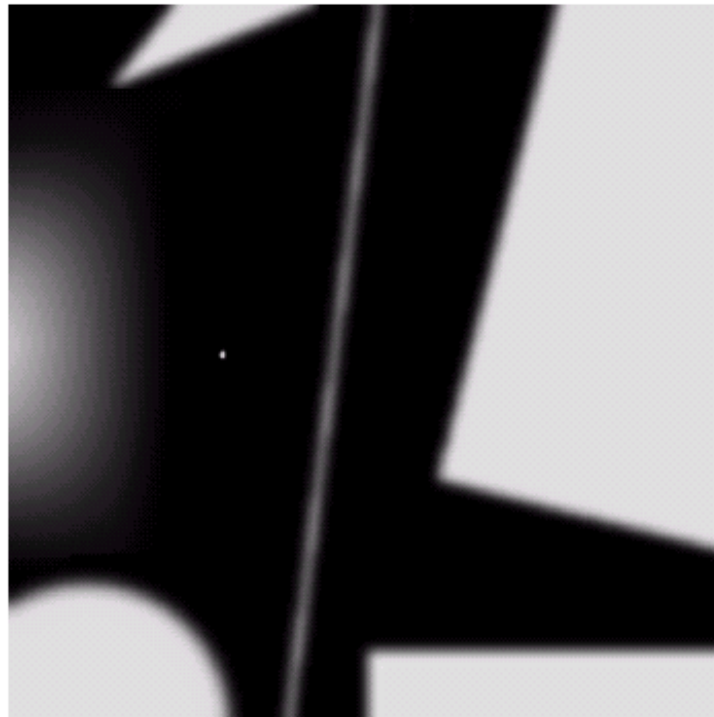
- ☐ Photo Enhancement
- ☐ Medical image visualization
- ☐ Industrial defect detection
- ☐ Electronic printing
- ☐ Autonomous guidance in military systems

Spatial Differentiation

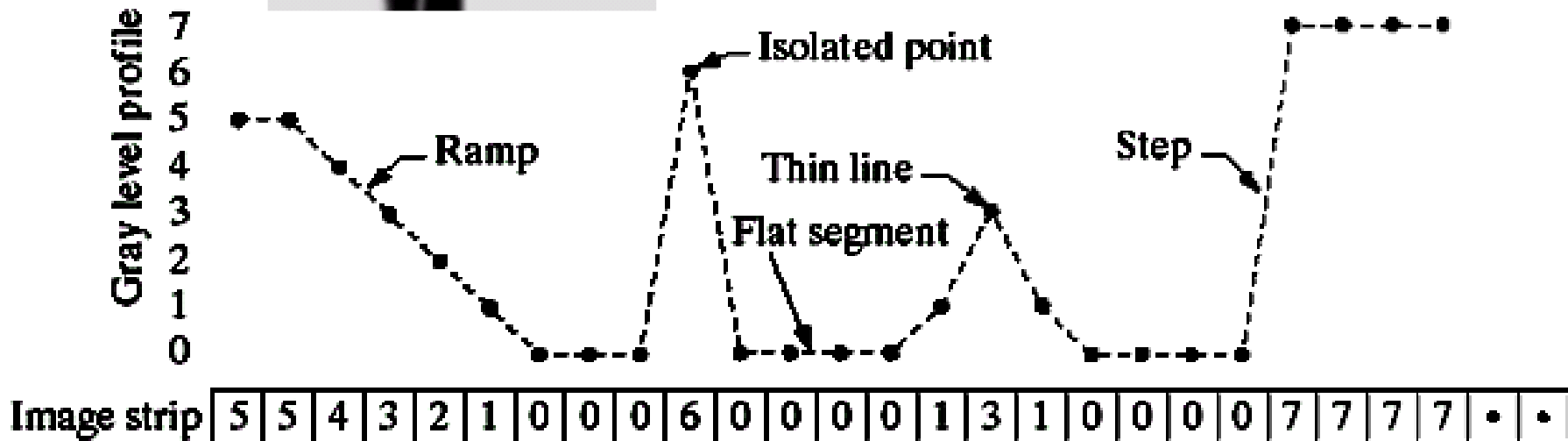
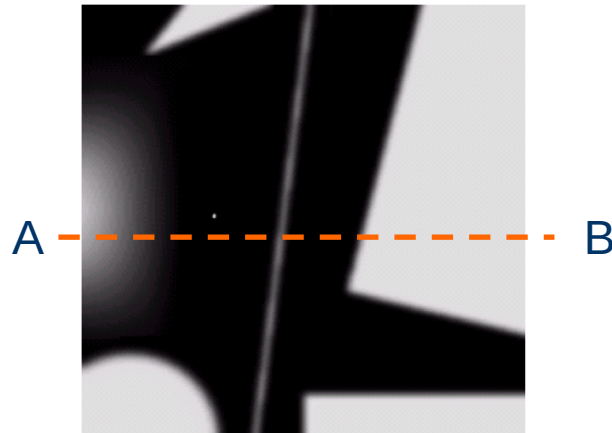
- Sharpening filters are based on **first- and second-order derivatives of image.**
- The derivatives of a digital function are defined **in terms of differences.**
- ***i.e.*** Sharpening filters are based on ***spatial differentiation***
- Differentiation measures **the rate of change of a function.**

Spatial Differentiation

- ◆ Let's consider a simple 1 dimensional example



Spatial Differentiation



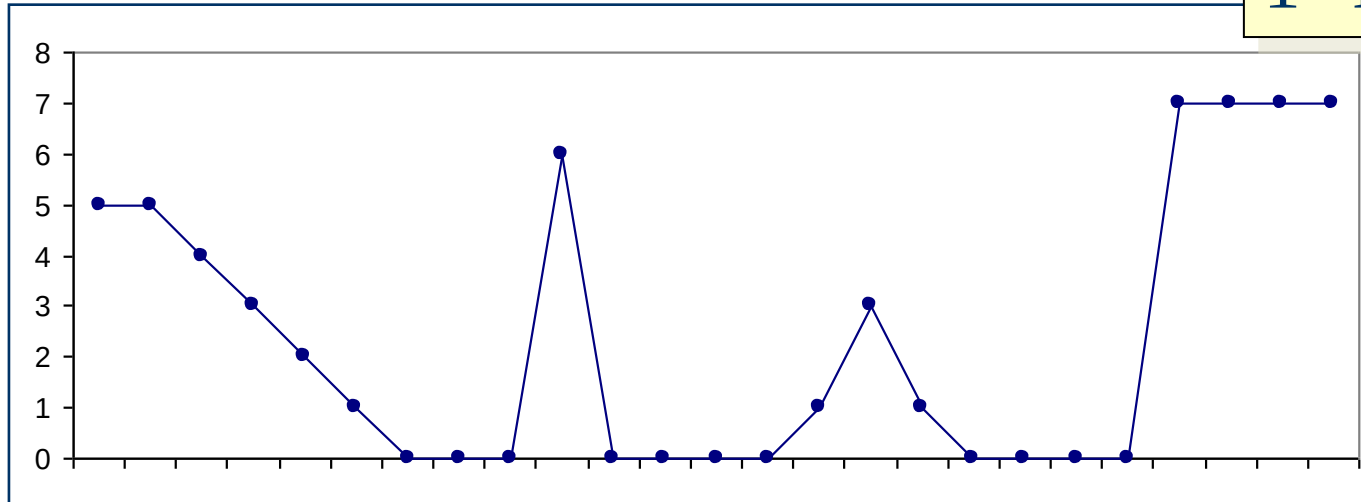
1st Derivative in Digital Form

- The 1st derivative of a function is given by:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

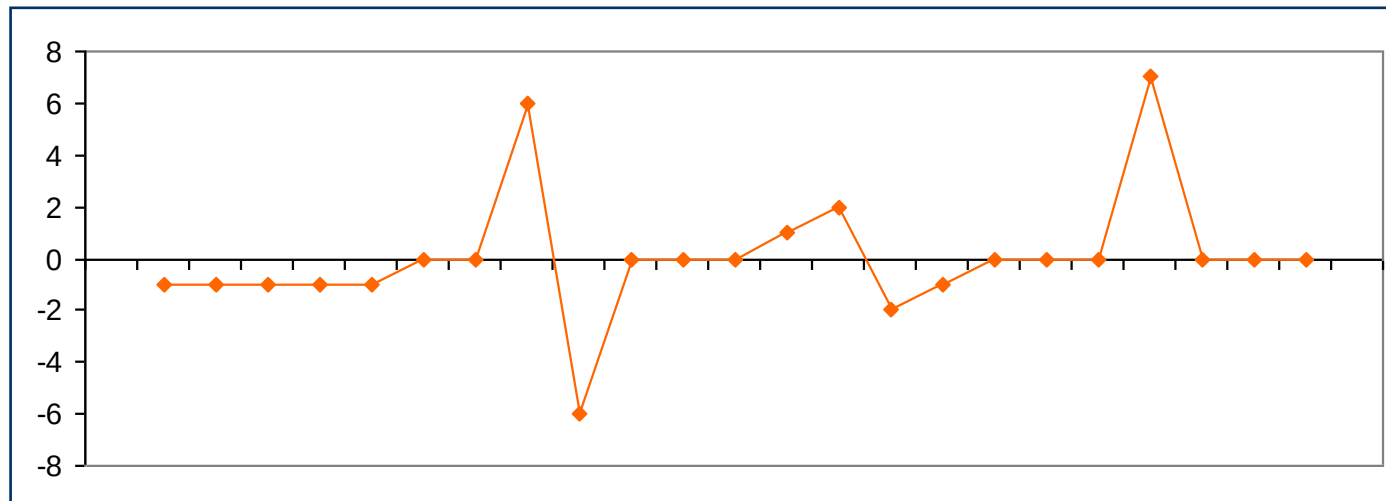
- Its just the difference between subsequent values and measures the rate of change of the function

1st Derivative



5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

	-1	-1	-1	-1	-1	0	0	6	-6	0	0	0	1	2	-2	-1	0	0	0	0	7	0	0	0	
--	----	----	----	----	----	---	---	---	----	---	---	---	---	---	----	----	---	---	---	---	---	---	---	---	--



Derivative is nonzero along the entire ramp, zero in flat area,.

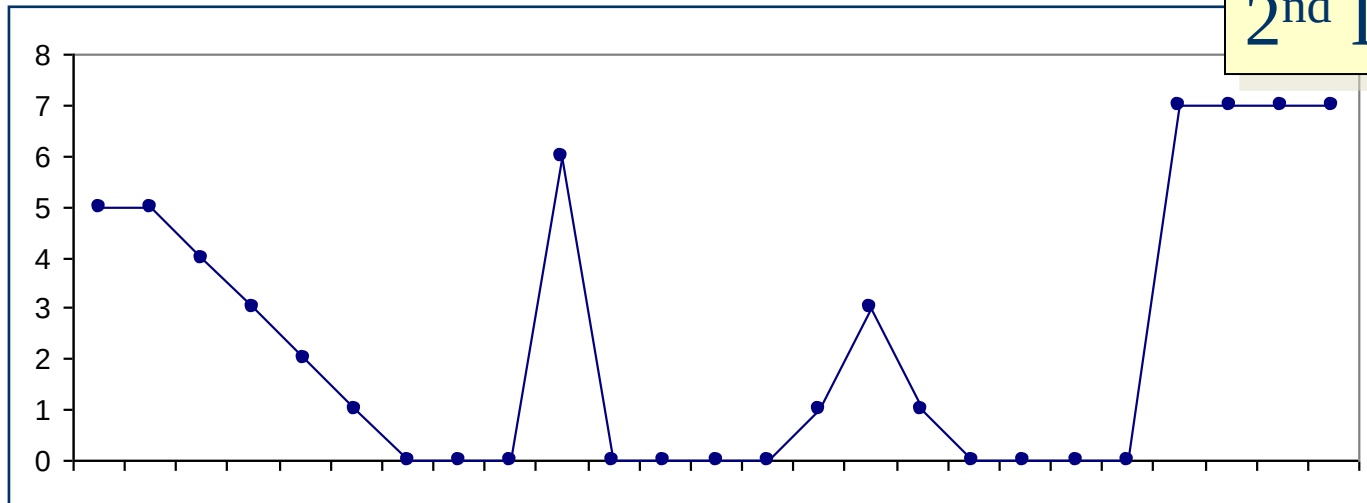
2nd Derivative in Digital Form

The 2nd derivative of a function is given by:

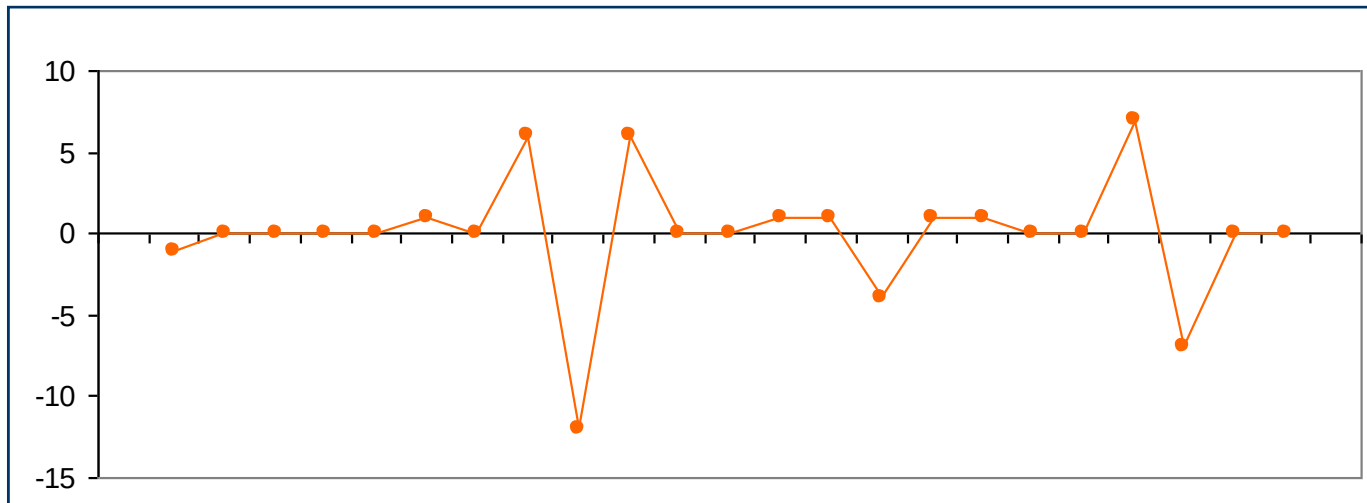
$$\frac{\partial^2 f}{\partial^2 x} = f(x+1) + f(x-1) - 2f(x)$$

Simply takes into account the values both before (backward) and after (forward) the current value

2nd Derivative



5	5	4	3	2	1	0	0	0	6	0	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7
-1	0	0	0	0	0	1	0	6	-12	6	0	0	1	1	-4	1	1	0	0	7	-7	0	0		



Derivative is nonzero at the onset and end of ramp, stronger response at and around the point.

Sharpening Spatial Filters

1. LAPLACIAN

- Use of 2nd Derivative for Image Enhancement

2. SOBEL (Gradient Operators)

- Use of 1st Derivative for Image Enhancement

Use of 2nd Derivative for Image Enhancement

The 2nd derivative is more useful for image enhancement than the 1st derivative - *Stronger response to fine detail*

The first sharpening filter we will look at is the ***Laplacian***

2nd derivatives for image Sharpening - For Two Dimensional

□ 2-D 2nd derivatives => Laplacian

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

=>discrete formulation

$$\begin{aligned}\nabla^2 f &= [f(x+1, y) + f(x-1, y) - 2f(x, y)] \\ &\quad + [f(x, y+1) + f(x, y-1) - 2f(x, y)] \\ &= [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)\end{aligned}$$

2nd Derivative in Two Dimension

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x) \quad \longrightarrow \quad \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$$

x kernel

$$\frac{\partial^2 f}{\partial y^2} = f(y+1) + f(y-1) - 2f(y) \quad \longrightarrow \quad \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

y kernel

0	0	0
1	-2	1
0	0	0

 +

0	1	0
0	-2	0
0	1	0

 =

0	1	0
1	-4	1
0	1	0

1. Laplacian Filter

So, the Laplacian can be given as follows:

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

We can implement it using this filter.

0	1	0
1	-4	1
0	1	0

Types of Laplacian Kernels

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a	b
c	d

FIGURE 3.39

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4).

(b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

Laplacian Image Enhancement

Another Example:

0	-1	0
-1	4	-1
0	-1	0



Original Image

+



**Laplace Sharpened
image**

=



**Laplace filtered
image**

Laplacian Filter

Example: apply the following Laplacian filter on the highlighted and underlined pixel

0	-1	0
-1	<u>4</u>	-1
0	-1	0

153	157	156	153	155
159	156	158	156	159
155	158	<u>154</u>	156	160
154	157	158	160	160
157	157	157	156	155

Step 1:

$$154 * 4 - 158 - 156 - 158 - 158 = -14$$

So the value after filter = **-14**

We call the resultant image: **sharpened image.**

Step 2:

Filtered image = original + sharpened image

The value in the filtered image = $154 - 14 = 130$

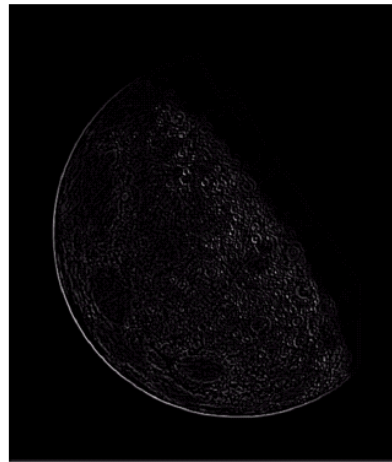
Laplacian Image Enhancement

0	1	0
1	-4	1
0	1	0



Original
Image

-



Laplacian
Filtered Image

=



Sharpened
Image

In the final sharpened image edges and fine detail are much more obvious

Simplified Image Enhancement

- The result of a Laplacian filtering is not an enhanced image.
- The entire enhancement can be combined into a **single filtering operation**

$$\begin{aligned} g(x, y) &= f(x, y) - \nabla^2 f \\ &= f(x, y) - [f(x+1, y) + f(x-1, y) \\ &\quad + f(x, y+1) + f(x, y-1) \\ &\quad - 4f(x, y)] \end{aligned}$$

0	1	0
1	-4	1
0	1	0

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f, & w_5 < 0 \\ f(x, y) + \nabla^2 f, & w_5 > 0 \end{cases}$$

Simplified Image Enhancement

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is positive.} \end{cases}$$

Simplified Image Enhancement

- ◆ The entire enhancement or sharpening can be done in one PASS.

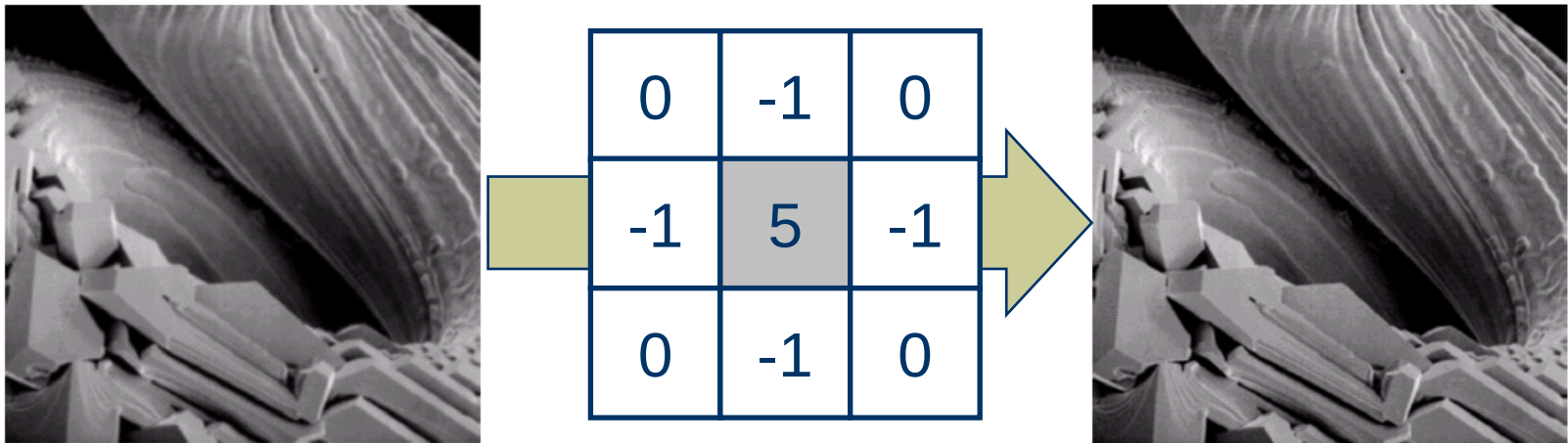
$$\begin{aligned} g(x, y) &= f(x, y) - \nabla^2 f \\ &= 5f(x, y) - f(x+1, y) - f(x-1, y) \\ &\quad - f(x, y+1) - f(x, y-1) \end{aligned}$$

0	-1	0
-1	5	-1
0	-1	0

We can implement it using this filter.

Simplified Image Enhancement

- ◆ This gives us a new filter which does the whole job for us in one step



Variants On The Simple Laplacian

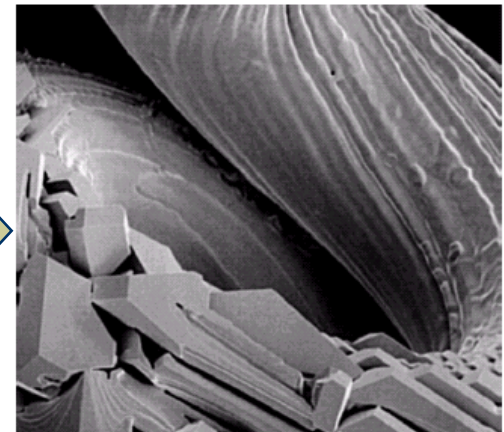
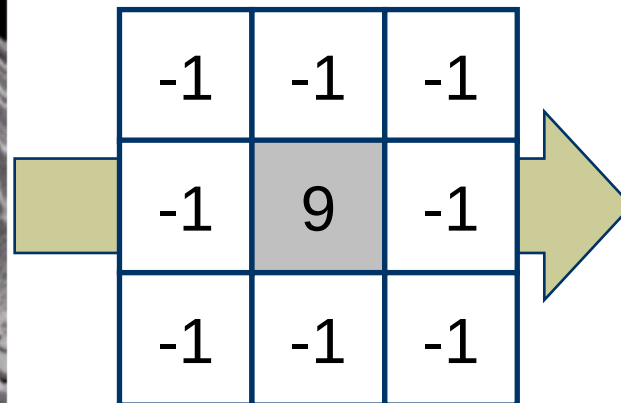
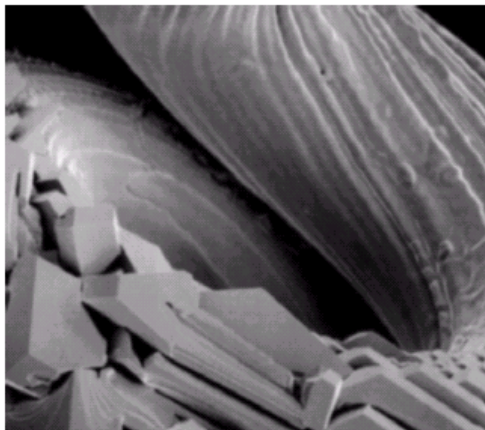
There are lots of slightly different versions of the Laplacian that can be used:

0	1	0
1	-4	1
0	1	0

**Simple
Laplacian**

1	1	1
1	-8	1
1	1	1

**Variant of
Laplacian**



Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

f

50	60	90	100
50	60	90	100
50	60	90	100
50	60	90	100

Laplacian

0	-1	0
-1	4	-1
0	-1	0

$\nabla^2 f$

-10	-20	20	10
-10	-20	20	10
-10	-20	20	10
-10	-20	20	10

$g = f + \nabla^2 f$

40	40	110	110
40	40	110	110
40	40	110	110
40	40	110	110

Composite
Laplacian

50	60	90	100
50	60	90	100
50	60	90	100
50	60	90	100

0	-1	0
-1	5	-1
0	-1	0

40	40	110	110
40	40	110	110
40	40	110	110
40	40	110	110

Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

	0	-1	0		
	-1	50	60	90	100
		4	-1		
	0	50	60	90	100
		-1	0		
		50	60	90	100
		50	60	90	100

$\nabla^2 f$

-10			

$g = f + \nabla^2 f$

40			

$$\begin{aligned}
 \nabla^2 f &= 4 * 50 - 50 - 50 - 60 - 50 \\
 &= 200 - 210 \\
 &= -10
 \end{aligned}$$

$$g(x, y) = 50 - 10$$

$$g(x, y) = 40$$



Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

	0	-1	0	
50	-1	4	-1	100
50	0	-1	0	100
50	60	90	100	
50	60	90	100	

$\nabla^2 f$

-10	-20		

$g = f + \nabla^2 f$

40	40		

$$\begin{aligned}
 \nabla^2 f &= 4 * 60 - 50 - 60 - 60 - 90 \\
 &= 240 - 260 \\
 &= -20
 \end{aligned}$$

$$g(x, y) = 60 - 20$$

$$g(x, y) = 40$$

Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

	0	-1	0
50	60 -1	90 4	100 -1
50	60 0	90 -1	100 0
50	60	90	100
50	60	90	100

$\nabla^2 f$

-10	-20	20	

$g = f + \nabla^2 f$

40	40	110	

$$\begin{aligned}
 \nabla^2 f &= 4 * 90 - 90 - 90 - 60 - 100 \\
 &= 360 - 340 \\
 &= 20
 \end{aligned}$$

$$g(x, y) = 90 + 20$$

$$g(x, y) = 110$$

Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

			0	-1	0
50	60	90	100		-1
50	60	90	100		0
50	60	90	100		
50	60	90	100		

$\nabla^2 f$			
-10	-20	20	10

$$g = f + \nabla^2 f$$

40	40	110	110

$$\begin{aligned}\nabla^2 f &= 4 * 100 - 100 - 100 - 100 - 90 \\ &= 400 - 390 \\ &= 10\end{aligned}$$

$$g(x, y) = 100 + 10$$

$$g(x, y) = 110$$

Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

	50	60	90	100
0	-1	0		
-1	50	60	90	100
	4	-1		
0	50	60	90	100
	-1	0		
	50	60	90	100

$$\nabla^2 f$$

-10	-20	20	10
-10			

$$g = f + \nabla^2 f$$

40	40	110	110
40			

$$\begin{aligned}\nabla^2 f &= 4 * 50 - 50 - 50 - 50 - 60 \\ &= 200 - 210 \\ &= -10\end{aligned}$$

$$g(x, y) = 50 - 10$$



$$g(x, y) = 40$$

Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

50	60	90	100
50	60	90	100
50	60	90	100
50	60	90	100

		0	-1	0
		-1	4	-1
		0	-1	0

$\nabla^2 f$

-10	-20	20	10
-10	-20	20	10
-10	-20	20	10
-10	-20	20	10

$g = f + \nabla^2 f$

40	40	110	110
40	40	110	110
40	40	110	110
40	40	110	110

$$\begin{aligned}
 \nabla^2 f &= 4 * 100 - 100 - 100 - 100 - 90 \\
 &= 400 - 390 \\
 &= 10
 \end{aligned}$$

$$g(x, y) = 100 + 10$$

$$g(x, y) = 110$$

Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

	0	-1	0	
	-1	50	60	90
		5	-1	100
	0	50	60	90
		-1	0	100
		50	60	90
		50	60	90

g

40			

$$\begin{aligned}g(x, y) &= 5 * 50 - 50 - 50 - 60 - 50 \\&= 250 - 210 \\&= 40\end{aligned}$$



Laplacian vs. Composite Laplacian

Example: Apply the Laplacian and composite Laplacian filters on the following blurred edge

50	60	90	100	
50	60	90	100	
50	60	90	100	
50	60	90	100	
		0	-1	0
		-1	5	-1
		0	-1	0

g

40	40	110	110
40	40	110	110
40	40	110	110
40	40	110	110

$$\begin{aligned}g(x, y) &= 5 * 100 - 100 - 100 - 100 - 90 \\&= 500 - 390 \\&= 110\end{aligned}$$

Use of 1st Derivatives for Image Enhancement

The another Sharpening Spatial filters is SOBEL (Gradient Operators).

What is Gradient of a Digital Image?

1st Derivatives of Enhancement – The Gradient

First derivatives in image processing are implemented using the magnitude of the gradient.

For a function $f(x, y)$, the gradient of f at coordinates (x, y) is defined as the 2-dimensional column vector:

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}.$$

$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$
$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$
$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$
$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$	$\begin{bmatrix} G_x \\ G_y \end{bmatrix}$

1st Derivatives of Enhancement – The Gradient

The magnitude of the vector is given by:

$$\begin{aligned}\nabla f &= \text{mag}(\nabla f) \\ &= [G_x^2 + G_y^2]^{1/2} \\ &= \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}.\end{aligned}$$

The magnitude of the gradient vector is often referred to
As the *gradient*.

Because of the computational burden of the above Eq.,
it is common Practice to approximate the magnitude of
the gradient using absolute values as :

$$\nabla f \approx |G_x| + |G_y|$$

This Eq. is simpler to compute.

The Gradient of a Digital Image

- The magnitude of this vector is given by

$$\text{mag}(\nabla f) = \sqrt{G_x^2 + G_y^2} \approx |G_x| + |G_y|$$

G_x

-1	1
----	---

This mask is simple, and no isotropic. Its result only horizontal and vertical.

G_y

1
-1

The Gradient – First-order Derivative

How can we compute first-order discrete image derivatives?

- There are various ways...
 - One dimensional forward differences
 - Roberts cross gradient operators
 - One dimensional central differences
 - Prewitt operators
 - Sobel operators

There is some debate as to how best to calculate these gradients.

Gradient Operators

Robert's Method

- The simplest approximations to a first-order derivative that satisfy the conditions stated in that section are

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

$$G_x = (z_9 - z_5) \text{ and } G_y = (z_8 - z_6)$$

$$\nabla f = \sqrt{(z_9 - z_5)^2 + (z_8 - z_6)^2}$$

$$\nabla f \approx |z_9 - z_5| + |z_8 - z_6|$$

Gradient Operators

- These mask are referred to as the Roberts cross-gradient operators.

-1	0
0	1

0	-1
1	0

Robert's Operators

Example: Apply the Robert's operators on the pixel Z_5

Z_1	Z_2	Z_3	
Z_4	Z_5 -1	Z_6 0	
Z_7	Z_8 0	Z_9 1	

Z_1	Z_2	Z_3	
Z_4	Z_5 0	Z_6 -1	
Z_7	Z_8 1	Z_9 0	

$$G_x = Z_5 * -1 + Z_6 * 0 + Z_8 * 0 + Z_9 * 1$$

$$G_x = Z_9 - Z_5$$

$$G_y = Z_5 * 0 + Z_6 * -1 + Z_8 * 1 + Z_9 * 0$$

$$G_y = Z_8 - Z_6$$

$$\text{Gradient} = |Z_9 - Z_5| + |Z_8 - Z_6|$$

Sharpening Spatial filters :

2) SOBEL (Gradient Operator)

- The Sobel operator provides differencing and smoothing effect of an image.
- Sobel operator consists of 3x3 convolution kernels. G_x is a simple kernel and G_y is rotated by 90°

-1	0	+1
-2	0	+2
-1	0	+1

G_x

+1	+2	+1
0	0	0
-1	-2	-1

G_y

Gradient Operators

Sobel Operator

$$\frac{\partial f}{\partial y} =$$

-1	-2	-1
0	0	0
1	2	1

Gy, Extract horizontal edges

$$\frac{\partial f}{\partial x} =$$

-1	0	1
-2	0	2
-1	0	1

Gx, Extract vertical edges

$$|G| = |Gx| + |Gy|$$

$$\nabla f \approx |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| \\ + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Pixel Arrangement

Sobel Operators

Example: Apply the Sobel operators on the pixel Z_5

Z_1 -1	Z_2 -2	Z_3 -1	
Z_4 0	Z_5 0	Z_6 0	
Z_7 1	Z_8 2	Z_9 1	

$$G_x = Z_1 * -1 + Z_2 * -2 + Z_3 * -1 + Z_4 * 0 + Z_5 * 0 + Z_6 * 0 + Z_7 * 1 + Z_8 * 2 + Z_9 * 1$$

$$G_x = -Z_1 - 2Z_2 - Z_3 + Z_7 + 2Z_8 + Z_9$$

$$G_x = (Z_7 + 2Z_8 + Z_9) - (Z_1 + 2Z_2 + Z_3)$$

Z_1 -1	Z_2 0	Z_3 1	
Z_4 -2	Z_5 0	Z_6 2	
Z_7 -1	Z_8 0	Z_9 1	

$$G_x = Z_1 * -1 + Z_4 * -2 + Z_7 * -1 + Z_2 * 0 + Z_5 * 0 + Z_8 * 0 + Z_3 * 1 + Z_6 * 2 + Z_9 * 1$$

$$G_x = -Z_1 - 2Z_4 - Z_7 + Z_3 + 2Z_6 + Z_9$$

$$G_x = (Z_3 + 2Z_6 + Z_9) - (Z_1 + 2Z_4 + Z_7)$$

$$\text{Gradient} = |(Z_7 + 2Z_8 + Z_9) - (Z_1 + 2Z_2 + Z_3)| + |(Z_3 + 2Z_6 + Z_9) - (Z_1 + 2Z_4 + Z_7)|$$

Example: Find the gradient of the following shaded pixels:

			P2	P4	
P1	10	10	10	200	200
	10	10	10	200	200
P3	10	10	10	200	200
	100	100	100	20	20
	100	100	100	20	20
	100	100	100	20	20

Gx			Gy		
-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

$$G_x(P1) = -10 - 20 - 10 + 0 + 0 + 0 + 10 + 20 + 10 = 0$$

10	10	10	200	200	200
-1	-2	-1			
10	10	10	200	200	200
0	0	0			
10	10	10	200	200	200
1	2	1			
100	100	100	20	20	20
100	100	100	20	20	20
100	100	100	20	20	20

10	10	10	200	200	200
-1	0	1			
10	10	10	200	200	200
-2	0	2			
10	10	10	200	200	200
-1	0	1			
100	100	100	20	20	20
100	100	100	20	20	20
100	100	100	20	20	20

$$G_y(P1) = -10 + 0 + 10 - 20 + 0 + 20 - 10 + 0 + 10 = 0$$

$$\nabla(P1) = |G_x| + |G_y|$$

$$\nabla(P1) = 0$$

The gradient is zero because P1 is located within a constant area

Example: Find the gradient of the following shaded pixels:

Gx(P2)

10	10 ₋₁	10 ₋₂	200 ₋₁	200	200
10	10 ₀	10 ₀	200 ₀	200	200
10	10 ₁	10 ₂	200 ₁	200	200
100	100	100	20	20	20
100	100	100	20	20	20
100	100	100	20	20	20

Gy(P2)

10	10 ₋₁	10 ₀	200 ₁	200	200
10	10 ₋₂	10 ₀	200 ₂	200	200
10	10 ₋₁	10 ₀	200 ₁	200	200
100	100	100	20	20	20
100	100	100	20	20	20
100	100	100	20	20	20

$$G_x(p2) = -10 - 20 - 200 + 0 + 0 + 0 + 10 + 20 + 200 = 0$$

$$G_y(p2) = -10 + 0 + 200 - 20 + 0 + 400 - 10 + 0 + 200 = 760$$

$$\nabla(P2) = |G_x| + |G_y|$$

$$\nabla(P2) = 0 + 760 = 760$$

The G_y is non-zero because P2 is located on a vertical edge

Example: Find the gradient of the following shaded pixels:

Gx(P3)

10	10	10	200	200	200
10	10	10	200	200	200
-1	-2	-1			
10	10	10	200	200	200
0	0	0			
100	100	100	20	20	20
1	2	1			
100	100	100	20	20	20
100	100	100	20	20	20

Gy(P3)

10	10	10	200	200	200
10	10	10	200	200	200
-1	0	1			
10	10	10	200	200	200
-2	0	2			
100	100	100	20	20	20
-1	0	1			
100	100	100	20	20	20
100	100	100	20	20	20

$$G_x(P3) = -10 - 20 - 10 + 0 + 0 + 0 + 100 + 200 + 100 = 360$$

$$G_y(P3) = -10 + 0 + 10 - 20 + 0 + 20 - 100 + 0 + 100 = 0$$

$$\nabla(P3) = |G_x| + |G_y|$$

$$\nabla(P3) = 0 + 360 = 360$$

The Gx here is non-zero because P3 is located on a horizontal edge

Example: Find the gradient of the following shaded pixels:

Gx(P4)

10	10	10	200	200	200
10	10	10	200	200	200
10	10	10	200	200	200
100	100	100	20	20	20
100	100	100	20	20	20
100	100	100	20	20	20

Gy(P4)

10	10	10	200	200	200
10	10	10	200	200	200
10	10	10	200	200	200
100	100	100	20	20	20
100	100	100	20	20	20
100	100	100	20	20	20

$$G_x(P4) = -10 - 20 - 200 + 0 + 0 + 0 + 100 + 200 + 20 = 90$$

$$G_y(P4) = -10 + 0 + 200 - 20 + 0 + 400 - 100 + 0 + 20 = 490$$

$$\begin{aligned} \nabla(P4) &= |G_x| + |G_y| \\ \nabla(P4) &= 90 + 490 = 580 \end{aligned}$$

The G_x and G_y here are non-zero because P4 is located on a horizontal and vertical edges

Gradient Operators

Prewitt Operator

□ is used for detecting edges horizontally and vertically.

$$|G| = |G_x| + |G_y|$$

$$\nabla f \approx |(z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)| \\ + |(z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)|$$

$$\frac{\partial f}{\partial y} = \begin{array}{|c|c|c|} \hline -1 & -1 & -1 \\ \hline 0 & 0 & 0 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Extract horizontal edges

G_y

$$\frac{\partial f}{\partial x} = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$$

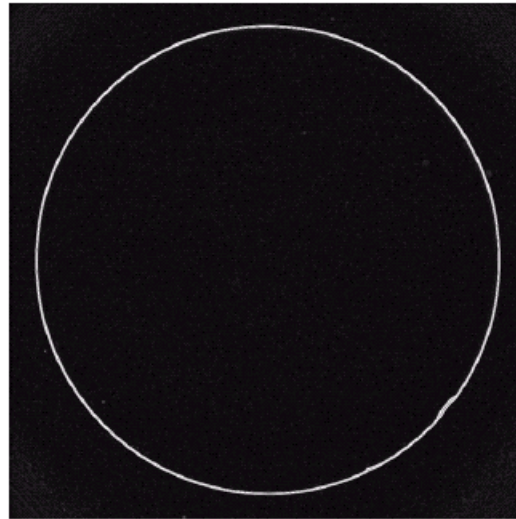
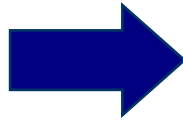
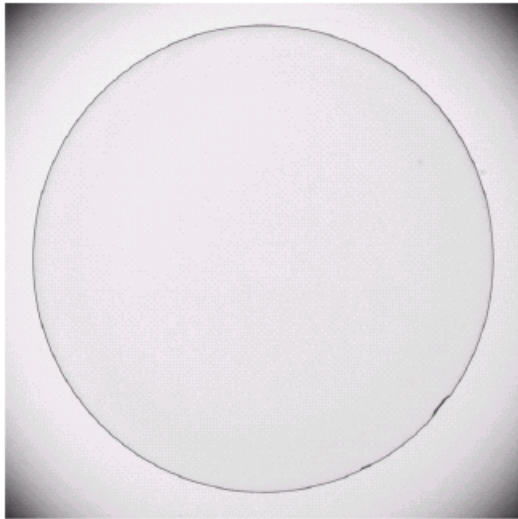
Extract vertical edges

G_x

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

Pixel Arrangement

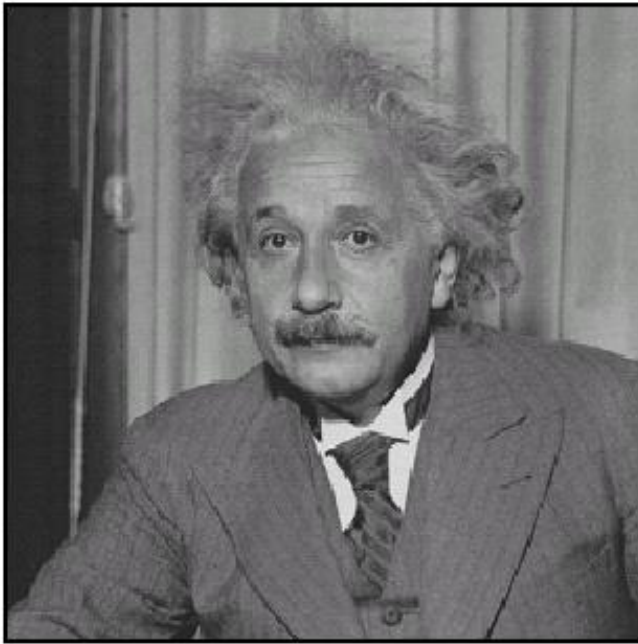
Sobel Operator: Example



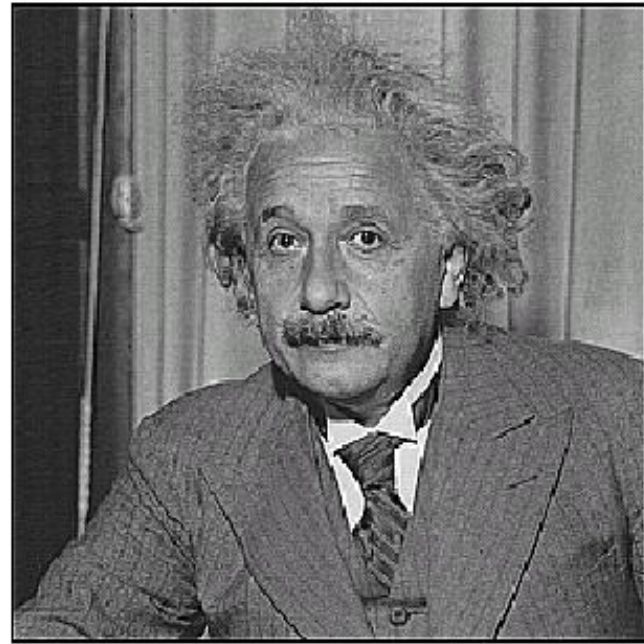
An image of a contact lens which is enhanced in order to make defects more obvious

Sobel filters are typically used for edge detection

Sharpening with Sobel Operator

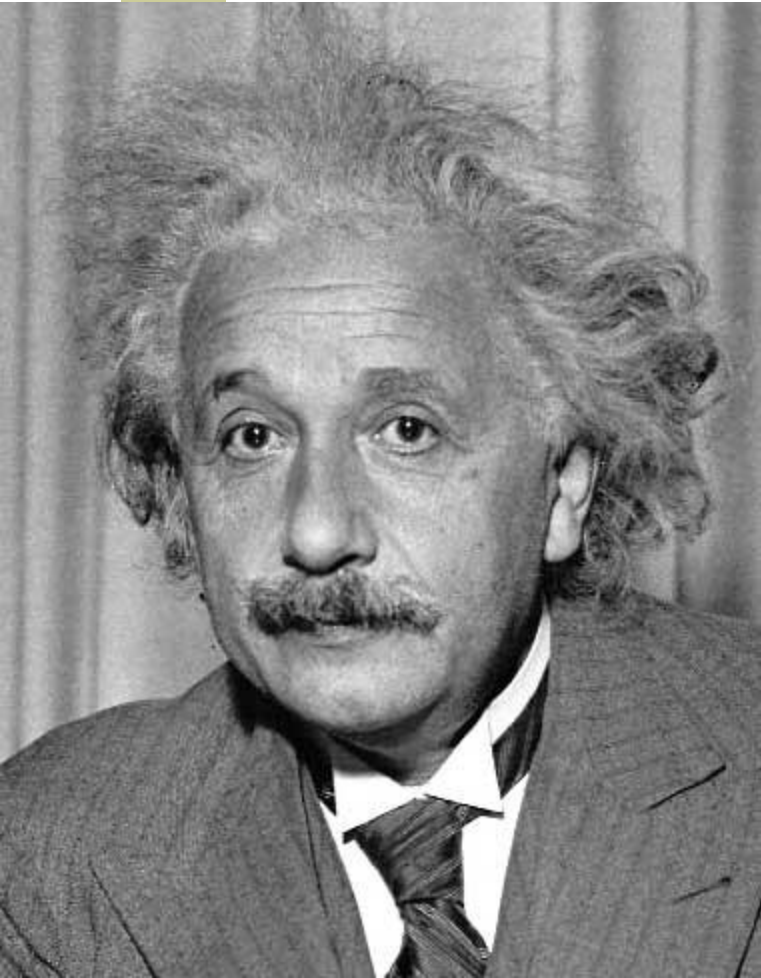


before



after

Sharpening with Sobel Operator



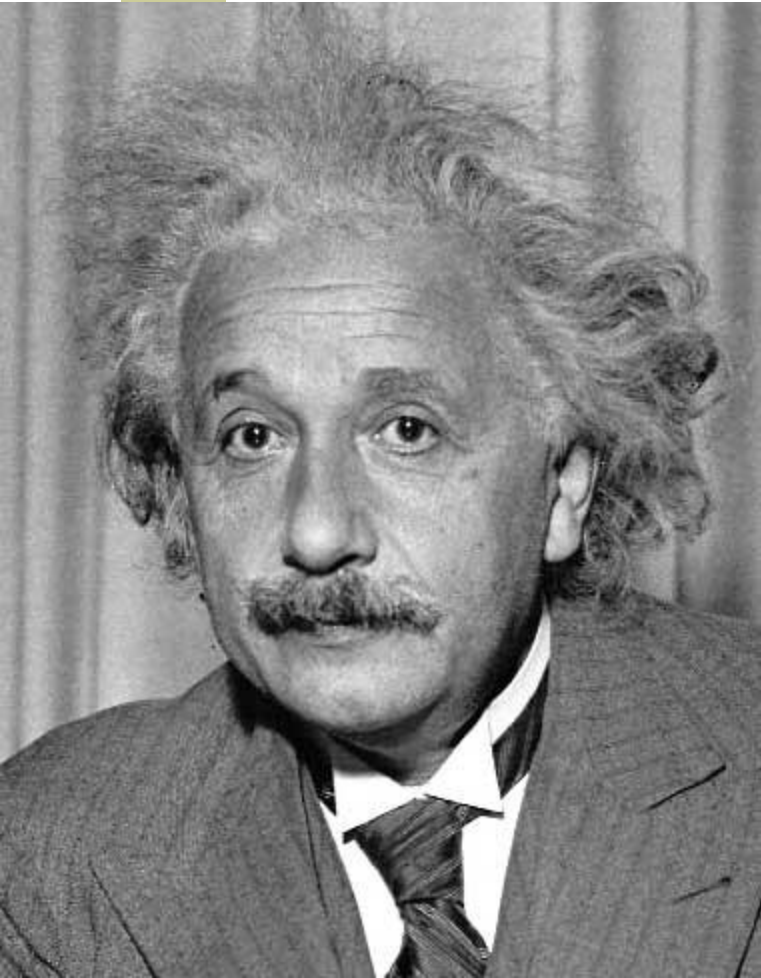
1	0	-1
2	0	-2
1	0	-1

Sobel



Vertical Edge
(absolute value)

Sharpening with Sobel Operator



1	2	1
0	0	0
-1	-2	-1

Sobel



Horizontal Edge
(absolute value)

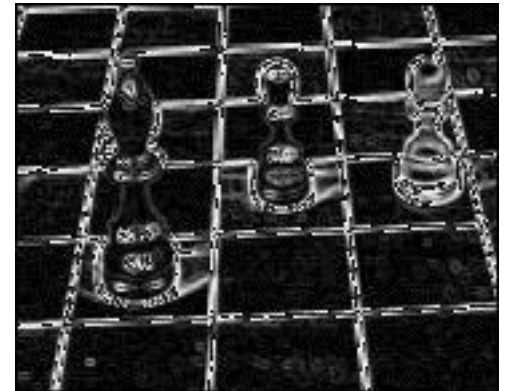
Sharpening Spatial Filters



Laplacian



Sobel



Difference Filter

- ❑ Also called as **Emboss filters**
- ❑ Enhances the details in the direction specific to the mask selected
- ❑ Four primary difference filter convolution masks, corresponding to the edges in the vertical, horizontal, and two diagonal directions are:

Vertical	Horizontal	Diagonal 1	Diagonal 2
$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

Difference Filter



Original image



Difference filtered image



Difference filtered image
added to the original image,
with contrast enhanced

Combining Spatial Enhancement Methods

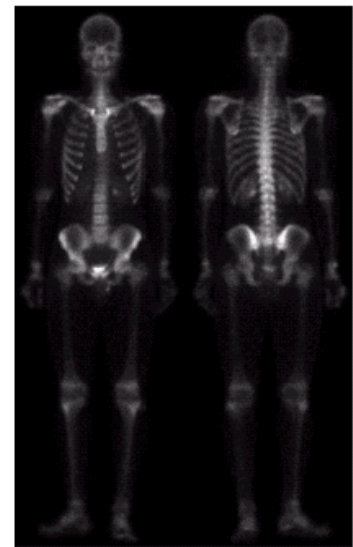
Successful image enhancement is typically not achieved using a single operation

Rather we combine a range of techniques in order to achieve a final result

This example will focus on enhancing the bone scan



Combining Spatial Enhancement Methods



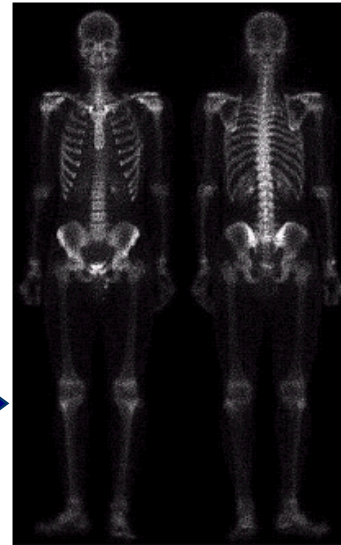
(a)

Laplacian filter of
bone scan (a)



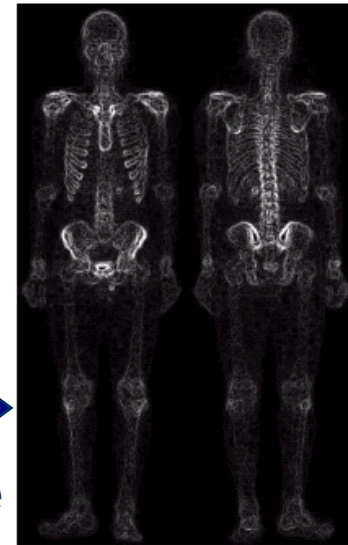
(b)

Sharpened version of
bone scan achieved
by subtracting (a)
and (b)



(c)

Sobel filter of bone
scan (a)



(d)

Combining Spatial Enhancement Methods

The product of (c) and (e) which will be used as a mask

(e)

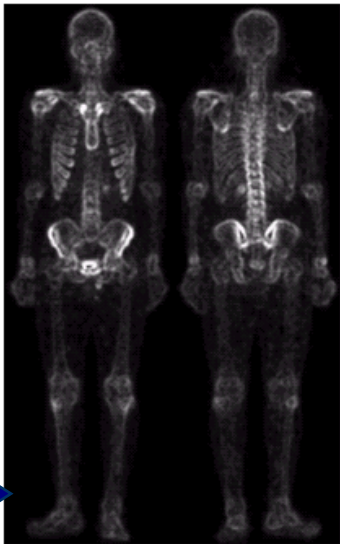
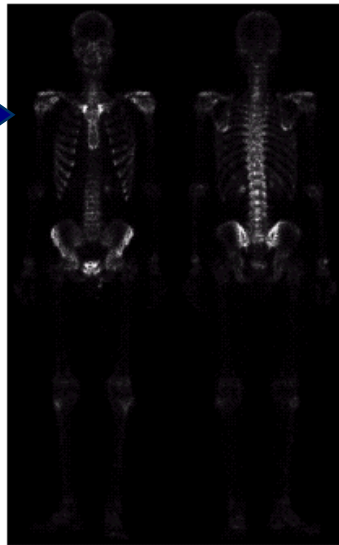


Image (d) smoothed with a 5*5 averaging filter

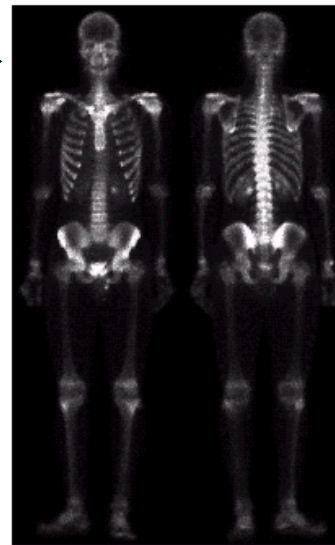
Sharpened image which is sum of (a) and (f)

(f)

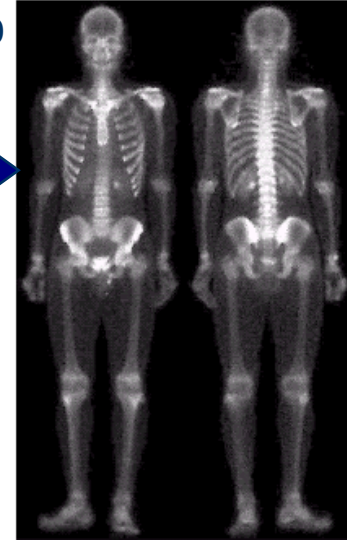


Result of applying a power-law trans. to (g)

(g)



(h)



Combining Spatial Enhancement Methods

Compare the original and final images

